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Nanowire Transistor Design Summary

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Device Design

This project presents an n-type gate-all-around silicon nanowire transistor simulated with the Silvaco Victory Device through nanoHUB. The device was modeled as a two-dimensional cylindrical structure. The radial direction represents the nanowire radius, while the vertical direction represents transport from source to drain.

The nanowire radius was set to 3 nanometers, giving the required 6-nanometer diameter. The gate length was set to 15 nanometers. The inner spacer length was set to 4 nanometers on each side of the gate. The supply voltage was 0.7 volts. The gate work function was 4.4 electron volts. The device temperature was 300 kelvin.

The channel used a light p-type background concentration of 1.0×10^{15} per cubic centimeter. The source and drain regions used n-type doping of 3.5×10^{20} per cubic centimeter. The extension regions used n-type doping of 8.0×10^{19} per cubic centimeter. The maximum doping concentration remained below the required limit of 5.0×10^{20} per cubic centimeter.

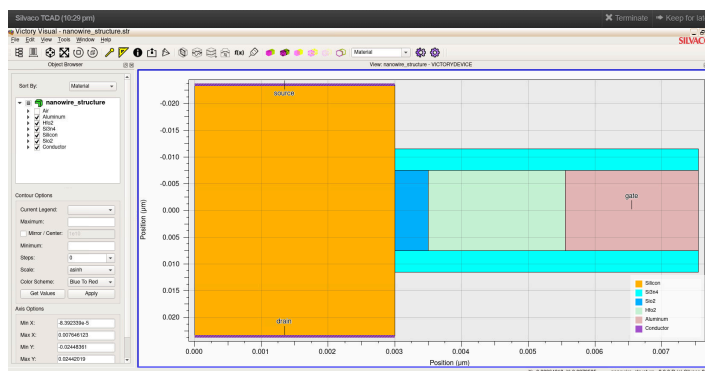


Figure 1. Simulated two-dimensional cylindrical nanowire structure showing the silicon channel, source, drain, inner spacers, silicon dioxide, hafnium dioxide, metal gate, and contacts.

Constraint Summary

The design meets the required 6-nanometer nanowire diameter, 15-nanometer gate length, 4-nanometer inner spacer length, 0.7-volt supply voltage, 300-kelvin temperature, 4.4-electron-volt gate work function, and maximum doping limit. The channel remains lightly doped, which helps improve electrostatic control in the short-channel nanowire structure.

Equivalent Oxide Thickness Calculation

The gate stack was designed to meet an equivalent oxide thickness of 0.9 nanometers. The stack uses 0.50 nanometers of silicon dioxide and 2.05 nanometers of hafnium dioxide.

Equivalent oxide thickness = silicon dioxide thickness + hafnium dioxide thickness $\times$ silicon dioxide permittivity / hafnium dioxide permittivity

Equivalent oxide thickness = $0.50 + 2.05 \times 3.9 / 20$

Equivalent oxide thickness = $0.50 + 0.39975$

Equivalent oxide thickness = 0.89975 nanometers

The final value is approximately 0.9 nanometers, so the gate stack satisfies the required equivalent oxide thickness constraint.

Simulation Models and Bias Conditions

The device was simulated using drift-diffusion transport with Fermi statistics. The simulation included Shockley-Read-Hall recombination, Auger recombination, doping-dependent mobility, field-dependent mobility, band gap narrowing, band-to-band tunneling, and Bohm quantum potential correction.

The main model line used in the final input deck was:

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models fermi srh auger conmob fldmob bgn bbt.std bqp.n
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The drain voltage was ramped from 0 volts to 0.7 volts. After the drain reached 0.7 volts, the gate voltage was swept from 0 volts to 0.7 volts. Smaller gate voltage steps were used near the upper voltage range to improve convergence. The final biased structure was saved at a drain voltage of 0.7 volts with a gate voltage of 0.7 volts.

Results

The drain current versus gate voltage curve shows normal n-type transistor behavior. The drain current is low at a gate voltage of 0 volts, then increases smoothly as the gate voltage rises. The curve remained stable after adding band-to-band tunneling and quantum correction.

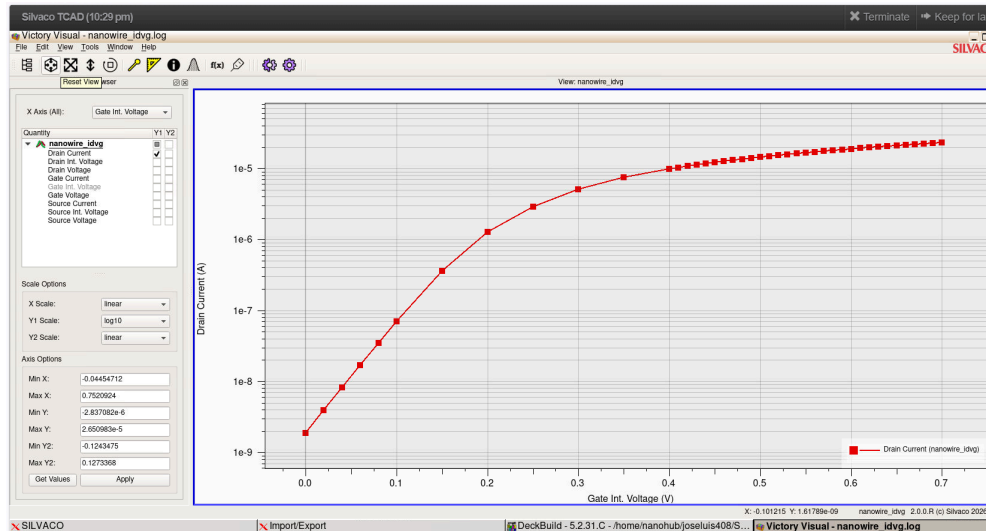


Figure 2. Drain current versus gate voltage at a drain voltage of 0.7 volts. The vertical axis uses a base-ten logarithmic scale.

The extracted approximate figures of merit are:

On current at a drain voltage of 0.7 volts and a gate voltage of 0.7 volts:
 2.33×10^{-5} amperes

Off current at a drain voltage of 0.7 volts and a gate voltage of 0 volts:
 about 2.0×10^{-9} amperes

On current divided by off current:
 about 1.0×10^4

Peak transconductance at a drain voltage of 0.7 volts:
 to be extracted from the final drain current versus gate voltage log

Maximum electron velocity at a drain voltage of 0.7 volts and a gate voltage of 0.7 volts:
 to be extracted from the final biased structure file

Conclusion

A two-dimensional cylindrical n-type silicon nanowire transistor was successfully created and simulated in the Silvaco Victory Device. The design meets the required geometry, voltage, work function, temperature, doping, and equivalent oxide thickness constraints. The final simulation includes mobility models, band-to-band tunneling, and quantum correction. The drain current versus gate voltage curve is stable, with an approximate on current of 2.33×10^{-5} amperes and an approximate off current of 2.0×10^{-9} amperes.